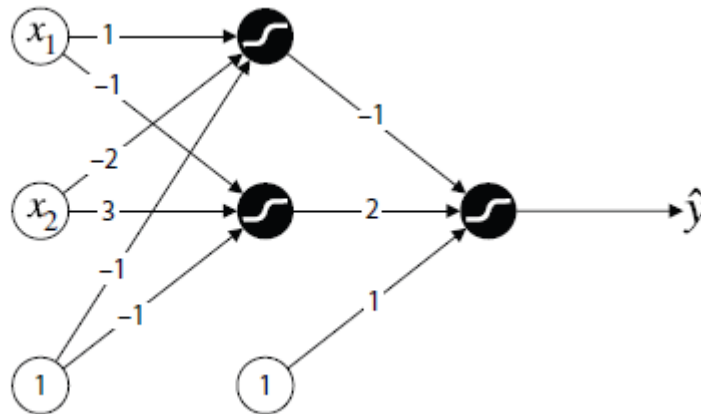


Exercise: Calculate value of neural network output represented by $\hat{y}$

The following image shows a neural network in which all the activations are sigmoid functions.

What would this neural network predict for the input (1, 1).



Solution on next page.

Solution

Let's call the outputs of the middle nodes h_1 and h_2 . These are calculated as follows:

$$h_1 = \sigma(1 \cdot x_1 - 2 \cdot x_2 - 1)$$

$$h_2 = \sigma(-1 \cdot x_1 + 3 \cdot x_2 - 1)$$

Plugging in $x_1 = 1$ and $x_2 = 1$, we get the following:

$$h_1 = \sigma(-2) = 0.119$$

$$h_2 = \sigma(1) = 0.731$$

The final layer is

$$\hat{y} = \sigma(-1 \cdot h_1 + 2 \cdot h_2 + 1).$$

Replacing the values previously obtained for h_1 and h_2 , we get

$$\hat{y} = \sigma(-0.119 + 2 \cdot 0.731 + 1) = \sigma(2.343) = 0.912.$$

Thus, the output of the neural network is 0.912.